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The Symbiotic X-ray Binary IGR J16194-2810: A Window on the Future Evolution of Wide Neutron Star Binaries From *Gaia*

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ABSTRACT

We present optical follow-up of IGR J16194-2810, a hard X-ray source discovered by the *INTEGRAL* mission. The optical counterpart is a $\sim 500\,L_\odot$ red giant at a distance of 2.1 kpc. We measured 17 radial velocities (RVs) of the giant over a period of 271 days. Fitting these RVs with a Keplerian model, we find an orbital period of $P_{\rm orb} = 192.73 \pm 0.01$ days and a companion mass function $f(M_2) = 0.365 \pm 0.003\,M_\odot$. We detect ellipsoidal variability with the same period in optical light curves from the ASAS-SN survey. Joint fitting of the RVs, light curves, and the broadband SED allows us to robustly constrain the masses of both components. We find a giant mass of $M_\star = 0.99^{+0.02}_{-0.03}\,M_\odot$ and a companion mass of $M_2 = 1.23^{+0.05}_{-0.03}\,M_\odot$, implying that the companion is a neutron star (NS). We recover a 4.06-hour period in the system's TESS light curve, which we tentatively associate with the NS spin period. The giant does not yet fill its Roche lobe, suggesting that current mass transfer is primarily via winds. MESA evolutionary models predict that the giant will overflow its Roche lobe in 5–10 Myr, eventually forming a recycled pulsar + white dwarf binary with a ~ 900 day period. IGR J16194-2810 provides a window on the future evolution of wide NS + main sequence binaries recently discovered via *Gaia* astrometry. As with those systems, the binary's formation history is uncertain. Before the formation of the NS, it likely survived a common envelope episode with a donor-to-accretor mass ratio $\gtrsim 10$ and emerged in a wide orbit. The NS likely formed with a weak kick ($v_{\rm kick} \lesssim 50$ km s^{-1}), as stronger kicks would have disrupted the orbit.

Keywords: Symbiotic binary stars (1674) — Low-mass X-ray binary stars (939) — Neutron stars (1108)

1. INTRODUCTION

Low-mass X-ray Binaries (LMXBs) are interacting binaries containing a compact object (i.e. a neutron star or black hole) accreting from a low-mass donor. "Symbiotic" X-ray binaries (SyXBs) are LMXBs in which the donor star is a red giant. While ∼200 LMXBs are known in the Milky Way, only a handful of SyXBs have been identified (Yungelson et al. 2019; Bahramian & Degenaar 2023). To date, only two SyXBs have known orbital periods: GX 1+4 ($P_{\rm orb} = 1161$ d) and 4U 1700+24 ($P_{\rm orb} = 4391$ d) (Hinkle et al. 2006, 2019; Yungelson et al. 2019). All SyXBs discovered so far are suspected to host wind-accreting neutron stars (NSs) based on the combination of their distinct outburst activity and relatively low average X-ray luminosities ($L_X \sim 10^{32}$ – 10^{36} erg s^{-1}) compared to LMXBs in which accretion occurs via Roche-lobe overflow (Yungelson et al. 2019). No SyXBs hosting black holes (BHs) have been discovered to date. We summarize the orbital periods and distances of the six known SyXBs in Table 1.

The wide orbits of SyXBs present a challenge for binary evolution modeling, because these systems likely survived a common envelope episode — in which the donor was $\gtrsim 10$ times more massive than the accretor — without suffering dramatic orbital shrinkage. Furthermore, most NSs are expected to experience a natal kick during supernova, with a typical velocity of ~ 300 km s^{-1} (Hobbs et al. 2005). In most cases, this kick (combined with the mass loss resulting from the supernova) is expected to unbind would-be SyXBs altogether (Hills 1983; Brandt & Podsiadlowski 1995).

Astrometric orbital solutions from the *Gaia* mission (Gaia Collaboration et al. 2023) have enabled the discovery of three BHs (El-Badry et al. 2023a; Chakrabarti et al. 2023; El-Badry et al. 2023b; Gaia Collaboration et al. 2024) and 21 NSs (El-Badry et al. 2024a,b) orbiting low-mass stars in unexpectedly wide (i.e. au-scale) orbits. Since these binaries will eventually evolve into SyXBs (Rodriguez et al. 2023), the challenge of explaining SyXB formation is closely related to the challenge of explaining the characteristics of the population of astrometric compact object binaries discovered by *Gaia*.

In this work, we carry out a thorough analysis of the SyXB IGR J16194-2810 in the optical. While the system has been studied extensively at X-ray wavelengths (Masetti et al. 2007; Ratti et al. 2010; Kitamura et al. 2014; Bozzo et al. 2024), the optical counterpart had, until recently, not been studied in detail. Shortly before this paper was completed, Hinkle et al. (2024) reported optical and near-infrared observations

of the source. Our analysis here is independent of theirs, but we compare to their work throughout the text. We synthesize archival photometry with new high-resolution optical spectroscopy, light curves, and multi-epoch radial velocities (RVs) to constrain the orbital parameters of the system and derive the stellar properties of the red giant companion.

The rest of this work is organized as follows. In Section 2, we discuss the discovery of IGR J16194-2810 and previous work characterizing the system. In Section 3, we provide an overview of the properties of the X-ray source and its optical counterpart. In Section 4, we present RV measurements of IGR J16194-2810 collected using the FEROS spectrograph over more than one orbit. In Section 5, we simultaneously fit the ASAS-SN light curves and observed RV curve of the luminous companion with a Keplerian orbit to derive the orbital period and neutron star mass. We also perform spectral energy distribution (SED) modeling of the system's multi-band photometry to study the properties of the giant star. In Section 6, we discuss the implications of our results for the formation of IGR J16194-2810. Finally, in Section 7, we summarize our main results and consider pathways for future work on SyXBs.

2. DISCOVERY AND PRIOR WORK

IGR J16194-2810 was discovered by *INTEGRAL* and reported in the second IBIS catalog by Bird et al. (2006). To improve the source localization, Masetti et al. (2007) performed follow-up observations with *Swift*/XRT. They confirmed the association of the X-ray source with a red giant (reporting a chance alignment probability $\sim 4 \times 10^{-5}$), and after obtaining a medium-resolution optical spectrum, characterized the optical counterpart ($V \sim 12.3$) as spectral type M2III. Since the X-ray light curve and spectrum of the source were typical of Galactic LMXBs, they classified IGR J16194-2810 as a new symbiotic X-ray binary hosting a NS.

Later on, Ratti et al. (2010) used *Chandra* X-ray observations to determine a precise localization of the X-ray source in IGR J16194-2810 to within 0.6 arcseconds. Further follow-up X-ray observations of the target were also obtained with *Suzaku* by Kitamura et al. (2014) and *Swift*/XRT and *NuSTAR* by Bozzo et al. (2024). While Kitamura et al. (2014) report the detection of both a soft and a hard emission component from their X-ray spectral analysis, Bozzo et al. (2024) demonstrate that a hot thermal blackbody component with $kT \simeq 1.10$ keV (completed by a power law at high energies) is sufficient to describe the broadband X-ray spectrum. Contrary to the earlier literature, Bozzo et al. (2024) are unable to derive a satisfactory spectral fit involving a Comptonization component, and rule out the presence of a cold thermal component arising from an accretion disk. While they demonstrate that the X-ray spectrum of IGR J16194-2810 is unlikely to undergo significant spectral variability over time, they also find that the source's X-ray brightness varies by a factor of 2–3 over few kilosecond timescales. Furthermore, they point out that the dynamic range of the X-ray luminosity is at least 180, as the source was significantly brighter in 2007 than it was in 2024.

Recently, Luna (2023) analyzed TESS (Ricker et al. 2015) observations of the red giant in IGR J16194-2810. Based on a Lomb-Scargle periodogram analysis (Lomb 1976; Scargle 1982) of the optical data, he reports a detection of a significant periodicity at a period of 242.839 minutes. He identifies this as the spin period of the neutron star in the SyXB, but finds that the period appears to be transient, since its significance varies both between and within TESS sectors. We consider this measurement suggestive but not definitive (see Section 5.7).

Kiraga (2012) analyzed ASAS (Pojmanski 2002) photometry of periodic variable stars coincident with objects from the ROSAT Bright Source Catalog (Voges et al. 1999). He detects a variability period of 192.8 days in the I-band light curve of IGR 16194-2810, tentatively attributing it to ellipsoidal modulation. We summarize the physical parameters measured so far in the literature for IGR J16194-2810 in Table 2.

As we were preparing this manuscript, an independent analysis of IGR J16194-2810 by Hinkle et al. (2024) appeared. The orbital period and most physical parameters we infer in our analysis are in good agreement with their findings. We discuss their work further in Section 7.1.

3. SOURCE PROPERTIES

We present a PanSTARRS (Chambers et al. 2016) g-band image ($50'' \times 50''$) centered on the optical coordinates of IGR J16194-2810 in the left panel of Figure 1. The image has been rescaled to enhance contrast and show the fainter stars surrounding the bright central star. We overlay the 0.6 arcsecond error circle corresponding to the *Chandra* X-ray localization of Ratti et al. (2010) in red.

To test the probability that the optical counterpart is a chance alignment with the X-ray source detected by *Chandra*, we use the third data release of the *Gaia* mission (DR3; Gaia Collaboration et al. 2023) to compute the projected number density of stars in a circle of radius 1° centered on the target. At each apparent magnitude threshold, we multiply the projected number density of stars at least as bright as this threshold by the projected area of the *Chandra* X-ray localization to derive a chance alignment probability. We find that, assuming an X-ray localization error of $0.6''$ based on the *Chandra* pointing uncertainty, the probability that a star with apparent G-band magnitude ≤ 11.4 would be a chance alignment is $\lesssim 3 \times 10^{-6}$. We plot the chance coincidence curve as a function of apparent G-band magnitude in the middle panel of Figure 1.

We display the position of the optical counterpart in IGR 16194-3810 on an extinction-corrected *Gaia* color-magnitude diagram in the right panel of Figure 1. For comparison, we also plot a sample of stars from *Gaia* DR3 that lie within 1 degree of the target and have well-constrained parallaxes (`parallax_over_error` > 3). We obtained

Table 1. A summary of known symbiotic X-ray binaries, with parameters taken from the literature. In the listed systems, the donor is on the red giant branch or asymptotic giant branch. We do not list systems with supergiant donors (e.g. 4U 1954+319), which would more properly be classified as high-mass X-ray binaries (see Hinkle et al. (2020) for a discussion). There are several other unconfirmed associations of red giants with X-ray sources that are not listed here (see Bahramian & Degenaar (2023) for a recent list of SyXB candidates).

SyXB	Orbital Period (days)	Distance (kpc)	Reference
(1)	(2)	(3)	(4)
GX 1+4	1161 ± 12	4.3	Hinkle et al. (2006)
IGR J17329-2731	?	$2.7^{+3.4}_{-1.2}$	Bozzo et al. (2018)
4U 1700+24	4391 ± 33	0.544 ± 0.010	Hinkle et al. (2019)
Sct X-1	?	$3.6^{+0.8}_{-0.7}$	De et al. (2022a)
SRGA J181414.6-225604	?	$14.6^{+2.9}_{-2.3}$	De et al. (2022b)
IGR J16194-2810	192.73 ± 0.01	2.129 ± 0.029	This Work

Table 2. Physical parameters of IGR J16194-2810 taken from the literature. The provided X-ray luminosities are based on the photogeometric distance and not corrected for absorption. We have applied the zeropoint correction of Lindegren et al. (2021) to the *Gaia* DR3 parallax.

Parameter	Description	Value	Reference
(1)	(2)	(3)	(4)
L_X	X-ray Luminosity (0.3–10 keV)	$0\text{–}3 \times 10^{34}$ erg s^{-1}	Bozzo et al. (2024)
	X-ray Luminosity (0.2–2.3 keV)	5.4×10^{32} erg s^{-1}	Merloni et al. (2024)
m_G	Apparent G-band Magnitude	11.421 ± 0.003	Gaia Collaboration et al. (2023)
α	Right Ascension (X-ray)	16h 19m 33.30 ± 0.05s	Ratti et al. (2010)
	Right Ascension (Optical)	16h 19m 33.34 ± 0.03s	Gaia Collaboration et al. (2023)
δ	Declination (X-ray)	$-28°$ 07' 40.3 ± 0.6"	Ratti et al. (2010)
	Declination (Optical)	$-28°$ 07' 39.90 ± 0.02"	Gaia Collaboration et al. (2023)
$\mu_\alpha \cos\delta$	Proper Motion in RA	-0.894 ± 0.043 mas yr^{-1}	Gaia Collaboration et al. (2023)
μ_δ	Proper Motion in Dec	-5.287 ± 0.027 mas yr^{-1}	Gaia Collaboration et al. (2023)
ϖ	Parallax	0.477 ± 0.036 mas	Gaia Collaboration et al. (2023)
d	Distance (Geometric)	$2.101^{+0.129}_{-0.151}$ kpc	Bailer-Jones et al. (2021)
	Distance (Photogeometric)	$2.131^{+0.160}_{-0.116}$ kpc	Bailer-Jones et al. (2021)
$P_{\rm spin}$	Spin Period	242.839 min	Luna (2023)

extinctions from the 3D dust map of Green et al. (2019). We filtered out most of the stars that lie below the main sequence using the prescription of Riello et al. (2021); these stars have anomalous color excess ratios due to contamination from crowding. Clearly, the optical counterpart lies high on the red giant branch.

4. DATA

4.1. *Photometry*

We retrieved archival photometry for the red giant in the V and g bands from the ASAS-SN survey (Shappee et al. 2014; Kochanek et al. 2017). The unphased ASAS-SN light curves are shown in the top panel of Figure 2. From a Lomb-Scargle periodogram analysis, we determine that the ASAS-SN light curves display approximately sinusoidal variability with dominant period $\approx$ 96 days. We interpret this as ellipsoidal modulation due to the tidal deformation of the red giant by the neutron star, implying that the true orbital period